



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

APRIL 2004

HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 2

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Hand in your answer booklets in 3 sections. Section A (Questions 1 - 3), Section B (Questions 4 - 6) and Section C (Questions 7 - 8).
- Start each **NEW** section in a separate answer booklet.

Total Marks - 87 Marks

- Attempt questions 1- 8
- All questions are NOT of equal value.

Examiner: *R. Boros*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 87
Attempt Questions 1 – 8
All questions are NOT of equal value

Answer each section in a SEPARATE writing booklet. Extra writing booklets are available.

SECTION A (Use a SEPARATE writing booklet)

Question 1 (11 marks)		Marks
(a)	Find $\int e^{\frac{x}{2}} dx$	1
(b)	Find the largest possible (natural) domain of the following function, $y = \log_e (\sin^{-1} x)$	2
(c)	Find a primitive function for	
(i)	$\frac{3x}{4+x^2}$	1
(ii)	$\frac{3}{4+x^2}$	2
(d)	Differentiate $y = \log_e (\sin^{-1} x)$	2
(e)	Solve $\tan \theta = \sin 2\theta$ for $0 < \theta < \pi$	3

Section A (continued)

Question 2 (13 marks)

Marks

(a) Show that $\tan 75^\circ = \frac{\sqrt{3}+1}{\sqrt{3}-1}$ 2

(b) (i) Show that $\frac{d}{d\theta}(\tan^3 \theta) = 3\sec^2 \theta(\sec^2 \theta - 1)$ 2

(ii) Hence, or otherwise, evaluate 3

$$\int_0^{\frac{\pi}{4}} \sec^4 \theta \, d\theta$$

(c) Show that $\frac{d}{dx}\left(\frac{\tan x}{e^{2x}}\right) = \left(\frac{\tan x - 1}{e^x}\right)^2$ 3

(d) Find the possible values of m if $y = e^{-mx}$ satisfies 3

$$\frac{d^2 y}{dx^2} + 2\frac{dy}{dx} - 15y = 0$$

Question 3 (12 marks)

(a) Find the exact value of 3

$$\int_{\frac{3\sqrt{3}}{2}}^3 \frac{2}{\sqrt{9-x^2}} \, dx$$

(b) Write down the general solution, in terms of π , of 4

$$2\sin \theta = \tan \theta$$

(c) Solve the equation 3

$$2\ln(3x+1) - \ln(x+1) = \ln(7x+4)$$

(d) Find $\int \frac{x+1}{x^2+2x-5} \, dx$ 2

SECTION B (Use a SEPARATE writing booklet)

Question 4 (10 marks)

Marks

- (a) State the domain of $y = 2 \sin^{-1} x$ 1
- (b) Prove that $\frac{d}{dx}(3^x) = 3^x \cdot \ln 3$ 2
- (c) Show that $\log_4 9 + \log_4 8 - 2 \log_4 6 = \frac{1}{2}$ 2
- (d) If $f(x) = x^2 + 2$ and $g(x) = 2x + 3$, find $f(g(x))$ 2
- (e) Differentiate with respect to x 3
- $$y = \ln \left(\frac{2x(x-1)^3}{\sqrt{x+1}} \right)$$

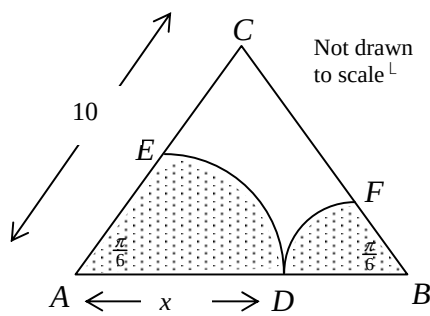
[Hint: Do not combine the answer as a single fraction]

Question 5 (13 marks)

- (a) Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x \cdot e^{\tan x} dx$ using the substitution $u = \tan x$. 3

Leave your answer in *exact* form.

- (b) Given $y = e^{\sin x}$, solve $\frac{d^2 y}{dx^2} - y = 0$ for $0 \leq x \leq 2\pi$ 4

- (c)  3
- $\triangle ABC$ is isosceles with $AC = BC = 10$,
 $\angle ABC = \angle CAB = \pi/6$,
 $AD = x$.
 AED and BDF are sectors of circles with radii AD and DB respectively.

- (i) Find an expression for BD . 3
- (ii) Show that the sum of the areas of the sectors AED and BDF is given by $\frac{\pi}{12} (2x^2 - 20\sqrt{3}x + 300)$ cm^2 3

SECTION B (continued)

Question 6 (12 marks)

Marks

- | | | | |
|-----|-------|--|---|
| (a) | (i) | Find $\int \cos^2 \theta d\theta$ | 2 |
| | (ii) | Hence, find $\int \frac{dx}{(1+x^2)^2}$ using the substitution $x = \tan \theta$ | 4 |
| (b) | (i) | Sketch the graph of the function $f(x) = e^x - 4$, showing clearly the coordinates of any points of intersection with the coordinate axes and the equations of any asymptotes. | 2 |
| | (ii) | On the same diagram, sketch the graph of the inverse function, $y = f^{-1}(x)$, showing clearly the coordinates of any points of intersection with the coordinate axes and the equations of any asymptotes. | 2 |
| | (iii) | Show that the x coordinate of any point of intersection of the graphs $y = f(x)$ and $y = f^{-1}(x)$ satisfies the equation $e^x - x - 4 = 0$. | 2 |

SECTION C (Use a SEPARATE writing booklet)

Question 7 (7 marks)

Marks

- (a) The region bounded by $y = \log_e x$, $x = 2$, $x = 5$ and the x axis is rotated about the x axis.

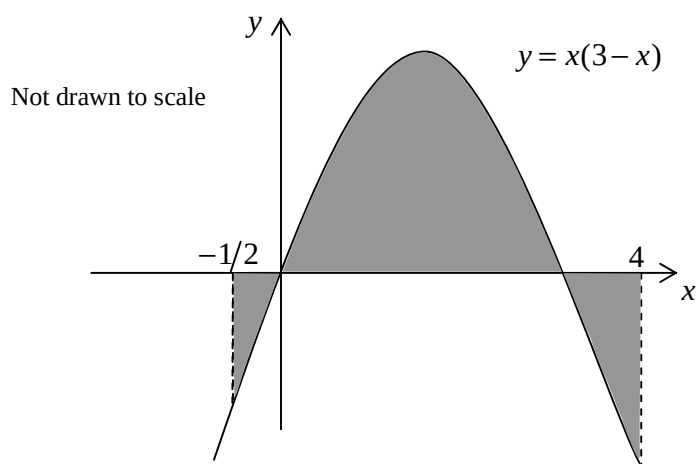
3

Use the Trapezoidal Rule with four function values to find an approximation to this volume.

Express your answer correct to 2 decimal places.

- (b)

4



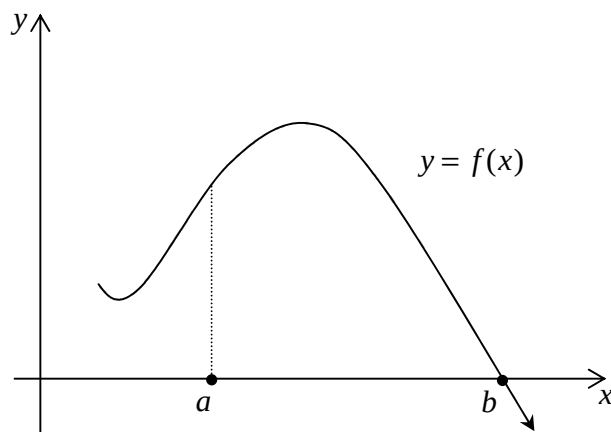
Find the size of the shaded area, correct to 2 decimal places.

SECTION C (continued)

Question 8 (9 marks)

Marks

(a)



Consider the above graph of $y = f(x)$. The value $x = a$ shown on the x axis is taken as the first approximation to the solution $x = b$ of $f(x) = 0$.

3

Is the second approximation obtained by Newton's Method a better approximation to b than the first approximation?

Justify your answer (using the diagram in your answer).

(b)

Consider $\tan^{-1} y = 2 \tan^{-1} x$

(i) Express y as a function of x , independent of any trigonometric ratio.

2

(ii) Show that the function has no turning points.

3

(iii) State the domain of the function.

1

THIS IS THE END OF THE PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$



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Mathematics Extension 1

Sample Solutions

Mathematics Extension 1
Assessment 2
Year 12 2004.

Question 1

(a) $2e^{\frac{\pi}{2}} + C$

(b) $\{x : 0 < x \leq 1\}$

(c) (i) $\frac{3}{2} \ln(4 + x^2)$

(ii) $\frac{3}{2} \tan^{-1} \frac{x}{2}$

(d) $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2} \sin^{-1} x}$

(e) $\tan \theta = \sin 2\theta \quad 0 < \theta < \pi$

$$\tan \theta - \sin 2\theta = 0$$

$$\frac{\sin \theta}{\cos \theta} - 2 \sin \theta \cos \theta = 0$$

$$\sin \theta (\sec \theta - 2 \cos \theta) = 0$$

$$\therefore \sin \theta = 0$$

$$\theta = 0, \pi$$

However $0 < \theta < \pi$.

$$\therefore \sec \theta - 2 \cos \theta = 0$$

$$1 - 2 \cos^2 \theta = 0$$

$$\cos^2 \theta = \frac{1}{2}$$

$$\cos \theta = \pm \frac{1}{\sqrt{2}}$$

$$\therefore \theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

Question 2

$$\begin{aligned} \text{(a)} \quad \tan 75^\circ &= \tan(30^\circ + 45^\circ) \\ &= \frac{\tan 30^\circ + \tan 45^\circ}{1 - \tan 30^\circ \tan 45^\circ} \\ &= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}} \cdot 1} \\ &= \frac{\frac{1 + \sqrt{3}}{\sqrt{3}}}{\frac{\sqrt{3} - 1}{\sqrt{3}}} \\ &= \frac{1 + \sqrt{3}}{\sqrt{3} - 1} \end{aligned}$$

$$\begin{aligned} \text{(b)(i)} \quad &\text{Let } u = \tan \theta \\ &\frac{du}{d\theta} = \sec^2 \theta \\ &y = u^3 \\ &\frac{dy}{du} = 3u^2 \\ &\frac{dy}{dx} = \frac{du}{d\theta} \cdot \frac{dy}{du} \\ &= \sec^2 \theta \cdot 3u^2 \\ &= 3 \sec^2 \theta \cdot \tan^2 \theta \\ &= 3 \sec^2 \theta (\sec^2 \theta - 1) \end{aligned}$$

$$\therefore \frac{d}{d\theta} \tan^3 \theta = 3 \sec^2 \theta (\sec^2 \theta - 1)$$

$$(ii) \quad \frac{d}{d\theta} \tan^3 \theta = 3 \sec^2 \theta (\sec^2 \theta - 1)$$

$$\therefore \int 3 \sec^4 \theta - 3 \sec^2 \theta \, d\theta = \tan^3 \theta + C$$

$$\int 3 \sec^4 \theta \, d\theta - \int 3 \sec^2 \theta \, d\theta = \tan^3 \theta + C$$

$$\int \sec^4 \theta \, d\theta - \int \sec^2 \theta \, d\theta = \frac{1}{3} \tan^3 \theta + C$$

$$\int \sec^4 \theta \, d\theta = \int \sec^2 \theta \, d\theta + \frac{1}{3} \tan^3 \theta + C$$

$$= \tan \theta + \frac{1}{3} \tan^3 \theta + C$$

$$\therefore \int_0^{\frac{\pi}{4}} \sec^4 \theta \, d\theta = \left[\tan \theta + \frac{1}{3} \tan^3 \theta \right]_0^{\frac{\pi}{4}}$$

$$= \left(\tan \frac{\pi}{4} + \frac{1}{3} \tan^3 \frac{\pi}{4} \right) - \left(\tan 1 + \frac{1}{3} \tan^3 1 \right)$$

$$= \frac{1}{3} + 1$$

$$= 1 \frac{1}{3}$$

(c) Let $u = \tan x$

$$\frac{du}{dx} = \sec^2 x$$

$$v = e^{2x}$$

$$\frac{dv}{dx} = 2e^{2x}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{v \cdot \frac{du}{dx} - u \cdot \frac{dv}{dx}}{v^2} \\&= \frac{(e^{2x})(\sec^2 x) - (\tan x)(2e^{2x})}{(e^{2x})^2} \\&= \frac{\sec^2 x - 2 \tan x}{e^{2x}} \\&= \frac{(\tan^2 x + 1) - 2 \tan x}{e^{2x}} \\&= \frac{\tan^2 x - 2 \tan x + 1}{(e^x)^2} \\&= \frac{(\tan x - 1)^2}{(e^x)^2} \\&= \left(\frac{\tan x - 1}{e^x} \right)^2\end{aligned}$$

(d)

$$y = e^{-mx}$$

$$\frac{dy}{dx} = -me^{-mx}$$

$$\frac{d^2y}{dx^2} = m^2 e^{-mx}$$

$$m^2 e^{-mx} + 2(-me^{-mx}) - 15e^{-mx} = 0$$

$$e^{-mx}(m^2 - 2m - 15) = 0$$

$$\therefore e^{-mx} = 0$$

no solution

$$\therefore m^2 - 2m - 15 = 0$$

$$(m - 5)(m + 3) = 0$$

$$m = 5, -3$$

Question 3

$$\begin{aligned} \text{(a)} \quad \int_{\frac{3\sqrt{3}}{2}}^3 \frac{2}{\sqrt{9-x^2}} dx &= 2 \int_{\frac{3\sqrt{3}}{2}}^3 \frac{1}{\sqrt{9-x^2}} dx \\ &= 2 \int_{\frac{3\sqrt{3}}{2}}^3 \frac{1}{\sqrt{3^2-x^2}} dx \\ &= 2 \left[\sin^{-1} \frac{x}{3} \right]_{\frac{3\sqrt{3}}{2}}^3 \\ &= 2 \left(\sin^{-1} \frac{3}{3} - \sin^{-1} \frac{3\sqrt{3}}{6} \right) \\ &= 2 \left(\sin^{-1} 1 - \sin^{-1} \frac{\sqrt{3}}{2} \right) \\ &= 2 \left(\frac{\pi}{2} - \frac{\pi}{3} \right) \\ &= \frac{\pi}{3} \end{aligned}$$

$$\text{(b)} \quad 2 \sin \theta = \tan \theta$$

$$2 \sin \theta = \frac{\sin \theta}{\cos \theta}$$

$$2 \sin \theta \cos \theta - \sin \theta = 0$$

$$\sin \theta (2 \cos \theta - 1) = 0$$

$$\therefore \sin \theta = 0$$

$$\theta = \pi n + (-1)^n \sin^{-1}(0), \quad n \in \mathbb{Z}$$

$$= \pi n + (-1)^n (0)$$

$$= \pi n$$

$$\therefore 2 \cos \theta - 1 = 0$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = 2\pi n \pm \cos^{-1}\left(\frac{1}{2}\right), \quad n \in \mathbb{Z}$$

$$= 2\pi n \pm \frac{\pi}{3}$$

$$(c) \quad 2\ln(3x+1) - \ln(x+1) = \ln(7x+4)$$

$$\frac{(3x+1)^2}{x+1} = 7x+4$$

$$9x^2 + 6x + 1 = (x+1)(7x+4)$$

$$9x^2 + 6x + 1 = 7x^2 + 11x + 4$$

$$2x^2 - 5x - 3 = 0$$

$$(2x+1)(x-3) = 0$$

$$\therefore 2x+1=0 \quad \text{or} \quad x-3=0$$

$$x = -\frac{1}{2} \quad \quad \quad x = 3$$

Since $x > 0$ a log of a negative is undefined, therefore $x=3$ is the only solution.

$$(d) \quad \frac{1}{2}\ln(x^2 + 2x - 5) + C$$

Q4

(a) $-1 \leq x \leq 1$

b) $y = 3^x$
 $x = \log_3 y$
 $= \frac{\log_e y}{\log_e 3}$

$\frac{dx}{dy} = \frac{1}{\ln 3 y}$

$\frac{dy}{dx} = \frac{d3^x}{dx} = 3^x \ln 3$

(c) $x = \log_4 \left(\frac{9x^2}{6^2} \right)$

$x = \log_4 2$

$4^x = 2$

$x = \frac{1}{2}$

(d) $f[g(x)] = (2x+3)^2 + 2$
 $= 4x^2 + 12x + 11$

(e) $y = \ln 2x + 3 \ln(x-1) - \frac{1}{2} \ln(x+1)$

$y' = \frac{1}{x} + \frac{3}{x-1} - \frac{1}{2(x+1)}$

25 (a) $\int_0^{\ln 4} \sec^2 x e^{\tan x} dx$
 $= \int_0^{\ln 4} \sec^2 x e^u \frac{du}{\sec^2 x}$
 $= [e^u]_0^{\ln 4}$
 $= e - 1$

$u = \tan x$
 $\frac{du}{dx} = \sec^2 x$
 $dx = \frac{du}{\sec^2 x}$
 $x = \frac{\pi}{4} \quad u = 1$
 $x = 0 \quad u = 0$

b) $y = e^{\sin x}$
 $y' = e^{\sin x} \cos x$
 $y'' = \cos x e^{\sin x} \cos x + e^{\sin x} (-\sin x)$
 $y'' - y = \cos^2 x e^{\sin x} - \sin x e^{\sin x} - e^{\sin x}$
 $e^{\sin x} (\cos^2 x - \sin x - 1) = 0$
 $1 - \sin^2 x - \sin x - 1 = 0 \quad (e^{\sin x} \neq 0)$
 $-\sin x (\sin x + 1) = 0$
 $3 \sin x = 0 \quad \text{or} \quad \sin x = -1$
 $x = 0, \pi, 2\pi \quad \text{or} \quad 3\pi/2$

Q5(c)

(i) In $\triangle ABC$ $\angle ACB = \frac{2\pi}{3}$

$AB^2 = 10^2 + 10^2 - 2 \times 100 \cos \frac{2\pi}{3}$ (Cos Rule)

$AB = 10\sqrt{3}$

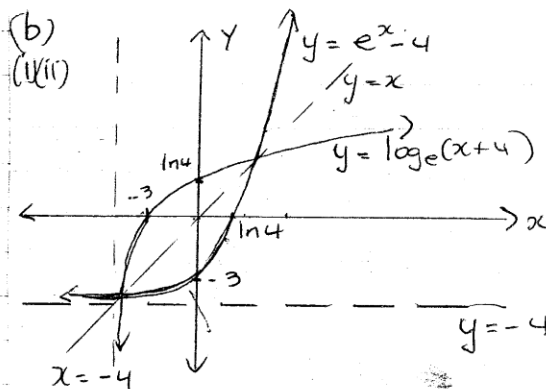
$DB = 10\sqrt{3} - x$

(ii) $\text{Area}(AED + BDF) = \frac{x^2 \pi}{2 \times 6} + \frac{1}{2} (10\sqrt{3} - x)^2 \times \frac{\pi}{6}$
 $= \frac{\pi}{2} (2x^2 + 300 - 20\sqrt{3}x)$

Q6 (a) (i) $\cos 2\theta = 2 \cos^2 \theta - 1$

$\int \cos^2 \theta d\theta = \frac{1}{2} \int (\cos 2\theta + 1) d\theta$
 $= \frac{1}{2} (\frac{1}{2} \sin 2\theta + \theta) + C$

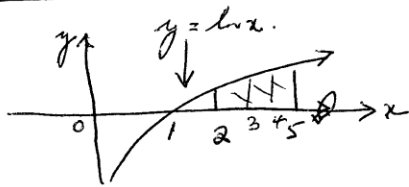
(ii) $\int \frac{dx}{(1+x^2)^2}$ $x = \tan \theta$
 $\frac{dx}{d\theta} = \sec^2 \theta$
 $dx = \sec^2 \theta d\theta$
 $= \int \frac{\sec^2 \theta d\theta}{(1+\tan^2 \theta)^2}$ $1 + \tan^2 \theta = \sec^2 \theta$
 $= \int \frac{\sec^2 \theta d\theta}{(\sec^2 \theta)^2}$
 $= \int \frac{1}{\sec^2 \theta} d\theta = \int \cos^2 \theta d\theta$
 $= \frac{1}{2} (\frac{1}{2} \sin 2\theta + \theta) + C$ (from (i))



(iii) The curves meet on the line $y = x$
 $e^x - 4 = x$
 $e^x - x - 4 = 0$

QUESTION 7

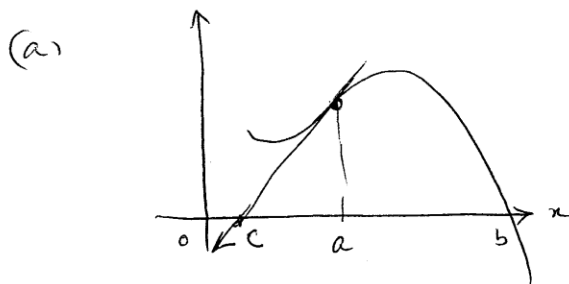
(a)



$$\begin{aligned}
 V &= \pi \int_1^5 (\ln x)^2 dx. \\
 &= \pi \cdot \frac{1}{2} \left[(\ln 2)^2 + (\ln 5)^2 + 2 [(\ln 3)^2 + (\ln 4)^2] \right] \\
 &= \underline{14.65 \cdot \pi^3} \quad (7.D.P.)
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) Shaded area} &= \left| \int_{-2}^0 x(3-x) dx \right| + \int_0^3 x(3-x) dx + \left| \int_3^4 x(3-x) dx \right| \\
 &= \left| \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_{-2}^0 \right| + \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_0^3 + \left| \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_3^4 \right| \\
 &= \left| 0 - \left(\frac{3 \cdot 4}{2} - \frac{1}{8} \right) \right| + \left(\frac{3 \cdot 9}{2} - \frac{27}{3} \right) + \left| \left(24 - \frac{64}{3} \right) - \left(\frac{3 \cdot 9}{2} - \frac{27}{3} \right) \right| \\
 &= \frac{5}{12} + \frac{9}{2} + \frac{11}{6} \\
 &= \frac{5 + 54 + 22}{12} \\
 &= \frac{81}{12} \\
 &= \underline{\left| \frac{27}{4} \pi^2 \right|} \approx \underline{6.75 \pi^2}
 \end{aligned}$$

QUESTION 8.



NO. The tangent to $y = f(x)$ where $x = a$ will cross the x-axis further from b than a, caused by the turning point between the root and the 1st approximation.

(b) (i) $\tan^{-1} y = 2 \tan^{-1} x$

$\therefore \tan(\tan^{-1} y) = \tan(2 \tan^{-1} x)$ *to think of.*

$$y = \frac{2 \tan(\tan^{-1} x)}{1 - \tan^2(\tan^{-1} x)}$$

$\therefore \boxed{y = \frac{2x}{1-x^2}}$ ✓✓

(3) ✓✓✓
must use a diagram.

(ii) now $\frac{dy}{dx} = \frac{(1-x^2) \cdot 2 - 2x \cdot 2x}{(1-x^2)^2}$
 $= \frac{2 - 2x^2 + 4x^2}{(1-x^2)^2}$
 $= \frac{2(1+x^2)}{(1-x^2)^2}$

Clearly for $\frac{dy}{dx} = 0$, $1+x^2 = 0$ which has no real solutions.

$\therefore \frac{dy}{dx} \neq 0 \therefore$ no turning points ✓✓✓

(iii) DOMAIN: $\boxed{\text{All real } \neq \pm 1.}$ ✓ (1)